Product Requirements Document - NC

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Executive Summary

Human relationships are breaking at the speed of biology — not at the speed of intention. A 40–80 millisecond amygdala threat-tagging gap causes the brain to classify the *tone* of an incoming message as a threat signal before the prefrontal cortex can process *communicative intent*. This is not a social media problem; it is a neuroscience problem. Social media spent 15 years amplifying that gap — training billions of people to scan for threat compulsively, at scale. The damage lands across every relationship that matters: couples, co-parents, colleagues, and friends.

NC (NeuroAI Cognitive Companion) is the counter-intervention. It is a multi-surface AI companion — browser extension and mobile app — that predicts what a person *actually means* and presents it to the receiver *before* the amygdala response can embed a false interpretation. NC is grounded in the neuroscience of mentalising (temporoparietal junction) and threat response (amygdala), deployed across digital text channels where ambiguity does its greatest relational damage.

Target users: Individuals experiencing relational friction in text-based digital communication — adults in intimate relationships, co-parenting arrangements, high-stakes professional relationships, and deep friendships — who are motivated to reduce misreading and restore signal fidelity in their most important conversations. **Neurodivergent users** (ADHD, ASD) are a named secondary segment: their intent-expression gap is wider than the neurotypical baseline and NC's person-aware model is particularly well-suited to bridging it. ND users are expected to be early adopters disproportionately represented in the initial cohort.

The problem NC solves: The intent gap — the systematic delta between what a sender means and what a receiver interprets — is invisible, pervasive, and cumulative. Each misread builds on the last. NC makes the gap visible and intervenes at the point of reception, before the narrative solidifies.

What Makes This Special

NC is not a communication coach, a sentiment analyser, or a conflict-resolution chatbot. It is a **foundation model of communicative intent** — the first product to combine three capabilities simultaneously:

1. **Person-aware intent modelling via the Soul Document.** NC conditions every intent prediction on *who the person actually is* — their ranked values, decision architecture,

and narrative history — not just what they wrote. The Soul Document is a portable, structured, model-agnostic personality representation (3-layer YAML schema) injected at inference time. This solves the cold-start problem from Day 1: new users receive accurate intent predictions without months of message history accumulation.

2. **Neuroscience-grounded inference, not pattern matching.** The core architecture models `P(intent | message, relational_history, soul_document, context)` — a probabilistic framework that explicitly separates the TPJ mentalising pathway from the amygdala threat pathway. No competitor has built intent prediction at this level of neurobiological precision.

3. **Personality sovereignty as a design constraint.** The Soul Document is local-first, AES-256 encrypted, not accessible to partners or third parties, and fully deletable. NC's differentiator is not data capture — it is data trust. Users who understand the alternative choose NC specifically because it doesn't monetise their personality.

Why users will choose NC over alternatives: Every other tool either analyses communication patterns without personal context (generic) or requires clinical intervention (inaccessible). NC is the only product that operates at consumer scale, in real time, across the channels where the intent gap actually lives — the text thread, the Slack message, the WhatsApp chain — with a personality layer that knows the person behind the words.

Core insight: Intent prediction accuracy is not a language problem. It is a person-knowledge problem. The same words, sent by different people in different relational contexts, mean entirely different things. NC is the first product built on this insight at production scale.

The flywheel: Science → Product → Data → Better Science. Every user interaction is a scientific data point. Every paper strengthens the product's credibility. Every credibility gain brings higher-quality users whose corrections improve the model. NC does not compete on data volume — it competes on data quality and scientific legitimacy. Where social media captured the amygdala, NC trains the prefrontal cortex — using the same neuroplasticity mechanisms, deliberately, for relational repair.

Project Classification

- **Project Type:** Multi-surface — browser extension (primary interception layer across web-based text channels) + mobile app (async support, Soul Document management, reflection layer)
- **Domain:** NeuroAI / Scientific Consumer — grounded in peer-reviewed neuroscience (amygdala threat response, TPJ mentalising, neuroplasticity), designed for everyday consumer deployment without clinical mediation

- **Complexity:** High — novel AI architecture (Soul Document + intent prediction model), dual-surface delivery, IRB-required research study embedded in Year 1, regulatory sensitivity around personal/relational data, and a scientific validity bar that most consumer products do not carry
- **Project Context:** Greenfield — no existing codebase, no legacy architecture, no inherited technical debt; all infrastructure, data pipelines, ML systems, and product surfaces built from first principles

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Success Criteria

> ***Note on dual-track success:*** *NC operates across two parallel tracks — a commercial product track and an embedded research track. This section defines* ***commercial product success only****. Research success (H1/H2/H3 experimental outcomes, annotation reliability, Brain-Score validation) is governed by the Soul Document Architecture Specification and the Experimental Plan in the product brief. These tracks are complementary but independently evaluated.*

User Success

NC succeeds for a user when it intervenes in the intent gap *before* a false interpretation becomes a relational narrative.

The success moment: A user reads an ambiguous message, sees NC's intent prediction, and modifies their response — editing, delaying, or discarding a reply that would have compounded misunderstanding. This is the primary behavioural signal of NC working. It does not require the user to report anything; it is measurable in-product.

The felt experience of success:

- *"I almost replied defensively — NC showed me he was actually nervous, not attacking."* (de-escalation before the reactive message sends)
- *"I didn't have to decode what she meant. NC told me, and she confirmed I got it right."* (accuracy + validation loop)
- *"We haven't had that kind of spiral in three weeks."* (downstream relational outcome — word-of-mouth signal, not primary metric)

User success is NOT:

- A user reading NC's prediction and ignoring it
- A user using NC on low-stakes, unambiguous messages
- A user who churns after the novelty wears off without behaviour change

Measurable user success criteria:

Metric	MVP Bar	Growth Bar
Reply modification rate (edit, delay, or discard after intent prediction surfaced)	≥ 20% of intent-prediction exposures	≥ 35%
Week-8 retention (user still active at 8 weeks — the neuroplasticity window)	≥ 50%	≥ 60%
Soul Document completion rate (users who complete intake and activate)	≥ 70% of signups	≥ 80%
User-reported accuracy ("NC got what they meant right") — in-app prompt, 5-point scale	≥ 3.5 / 5.0	≥ 4.0 / 5.0

Business Success

12-month commercial success (end of fellowship year):

NC has commercial legs if — independent of the research study — a cohort of real users is staying, engaging, and telling others. The following three signals together constitute a go/no-go signal for commercial investment:

Signal	Target	Why It Matters
Active non-study users on browser extension	300–500 by Month 12	Proves demand outside research recruitment
Week-8 retention	≥ 60%	Proves the neuroplasticity use-case — users come back because it's changing something
Institutional letter of intent	≥ 1 (couples therapist / HR platform / mediation service)	Proves professional market sees clinical/institutional value

What business success is NOT at 12 months:

- 10,000 downloads with 10% retention
- Revenue (pre-commercial; fellowship year is proof-of-concept)
- Viral growth (NC is intimate-use product; growth is trust-gated, not share-able)

18–24 month commercial horizon (post-fellowship):

Metric	Target
Paid subscriber conversion	≥ 15% of active free users
Monthly recurring revenue	£10K MRR as seed-round signal
Platform partnerships (therapy, HR, co-parenting apps)	2–3 signed integrations
Waitlist for mobile app (iOS)	2,000+ at launch

Technical Success

Intent prediction accuracy — the core technical bar:

NC must be *right often enough to be trusted*. Below the MVP floor, users experience NC as noise — it creates doubt without confidence and churns within days. Above the growth target, NC earns habitual use.

Threshold	Accuracy	Condition
MVP floor	**≥ 75%** on ambiguous-valence messages	Message + relational history, no Soul Document
Growth target	**≥ 85%**	Message + relational history + Soul Document active
Vision ceiling	**≥ 90%**	90+ days of relational history + full Soul Document

Rationale for 75% floor: Unaided human interpretation of emotionally ambiguous digital text is accurate approximately 50–60% of the time (informed by intent-gap annotation study baseline). NC at 75% is meaningfully better than unaided reading — sufficient for users to trust it. NC below 75% is not better enough to change behaviour.

Latency — the neuroscience constraint:

NC must surface the intent prediction *before* the amygdala response embeds a false interpretation. The relevant biological window is 40–800ms from stimulus receipt to cortical processing completion. For a browser extension intercepting an incoming message:

Target	Threshold

Intent prediction surface time	**< 800ms** from message receipt to UI display
Soul Document injection overhead	**< 200ms** additional (included in 800ms total)
Extension load time	**< 1s** on cold start

Privacy and security — hard constraints (not aspirational):

These are not technical stretch goals. They are product-defining constraints. Failure on any of these is product failure:

- Soul Document: local-first storage, AES-256 encryption at rest, device keychain gating
- No Soul Document content transmitted to server without explicit per-session user consent
- Full data deletion executable by user in ≤ 3 taps, with confirmation of deletion
- Partner access to another user's Soul Document: **architecturally prohibited in v1**
- GDPR Article 25 (privacy by design) compliance from Day 1 — not retrofitted

Cross-surface reliability:

Surface	Minimum supported channels (MVP)
Browser extension	Gmail, WhatsApp Web, Slack (web), iMessage web
Mobile app	iOS — Soul Document management + async reflection only (not real-time interception)

Measurable Outcomes Summary

Category	Metric	MVP Target	Growth Target
User	Reply modification rate	20%	35%
User	W8 retention	50%	60%
User	Soul Document activation	70%	80%
User	User-reported accuracy	3.5 / 5.0	4.0 / 5.0
Business	Active non-study users (Month 12)	300	500
Business	Institutional LOI	1	3
Technical	Intent accuracy (no Soul Doc)	75%	—

Technical	Intent accuracy (Soul Doc active)	—	85%
Technical	End-to-end latency	< 800ms	< 500ms
Technical	Privacy constraints	All met	All met

Product Scope

MVP — Minimum Viable Product

What must work for NC to be useful at all.

The MVP is the smallest version of NC that can demonstrate the intent gap intervention in a real user's real conversation, with sufficient accuracy to change behaviour and sufficient privacy to earn trust.

MVP includes:

- **Browser extension (Chrome MV3):** Real-time intent prediction on incoming messages across Gmail, WhatsApp Web, Slack (web), and iMessage web. Displays intent prediction as a non-intrusive overlay before the user replies.
- **Soul Document v1 intake:** Guided 3-question self-report (10 minutes). NC extracts and structures 3-layer YAML automatically. User reviews and activates before first use.
- **Intent gap visualiser:** Shows the predicted intent distribution (8-category label space) with confidence levels. High-entropy predictions display "high ambiguity" flag — no false confidence.
- **Reply modification UX:** Passive instrumentation of reply editing behaviour. No prompts or nudges — the prediction itself is the intervention.
- **Local-first data model:** Soul Document stored on-device, AES-256 encrypted, no cloud sync.
- **Full data deletion:** User can delete all NC data in ≤ 3 taps.
- **Mobile app (iOS + Android) — Soul Document layer only:** Cross-platform Soul Document management, intake, and async reflection prompts. Built in React Native or Flutter. No real-time message interception on mobile in MVP — for either platform. Android minimum supported version: API 26 (Android 8.0), covering ~95% of active Android devices. Soul Document encrypted at rest using Android Keystore (Android) and iOS Keychain (iOS); biometric/PIN gate on both platforms.

MVP explicitly excludes:

- Multi-user / partner consent networking (Horizon 2)
- Voice channel support

- Real-time message interception on mobile (iOS or Android) — Growth feature for both platforms
- Fine-tuned personalisation beyond Soul Document (requires message history accumulation)
- Revenue / payment infrastructure

Growth Features (Post-MVP)

What makes NC competitive and retentive.

- **Relational history deepening:** Progressive improvement in intent accuracy as message history accumulates. Accuracy trajectory surfaced to user ("NC knows you better now").
- **Pattern-level insights:** Weekly reflection on recurring intent gap patterns — "This phrasing is consistently misread between you and [X]."
- **Mobile real-time interception (iOS + Android):** Extend intent prediction to native mobile messaging — iOS (iMessage, WhatsApp iOS via share extension) and Android (WhatsApp, SMS via Accessibility Services or notification listener, pending platform policy review). iOS first given permissions complexity; Android to follow. Both require independent platform compliance review before shipping.
- **Soul Document v2:** Layers 4–6 (Narrative History, Relational Maps, Temporal Patterns). Richer cold-start accuracy. Automated story tagging replaces manual RA tagging.
- **Institutional integration API:** Headless NC engine consumable by couples therapy platforms, HR tools, and co-parenting apps.
- **Precision accuracy tooling:** In-app accuracy feedback loop — user can flag wrong predictions. Feeds anonymised improvement signal (with consent).

Vision (Future)

What NC becomes if the science holds and the product earns trust.

- **Consent network (Horizon 2):** Two users in a relationship can opt into mutual intent sharing — both see each other's intent predictions with explicit, revocable bilateral consent. Ethics review required; coercive control design constraint governs architecture.
- **Voice channel:** Real-time intent prediction on voice messages and calls. Requires new model architecture (prosodic + semantic fusion).
- **Neuroplasticity programme:** Structured 60-day programme grounded in H3 mechanism — designed to shift users from threat-first to intent-first interpretation as a lasting cognitive change, not just an in-the-moment tool.

- **Clinician dashboard:** Read-only, consent-gated view for therapists and mediators working with NC users — aggregate pattern trends, not message content.
- **Cross-language support:** Intent prediction beyond English, beginning with languages with the highest dyadic digital communication volume (Spanish, Mandarin, Hindi).
- **Foundation model open release:** The NC intent prediction model and annotated dataset released as open-source scientific infrastructure — the "ImageNet for communicative intent."

User Journeys

Journey 1: Core Receiver — Success Path

Persona: Maya, 34 — Partner in a 6-year relationship

> *"He's always so short with me when he's stressed. I never know if he's angry at me or just overwhelmed."*

Opening Scene

It's 6:43pm on a Wednesday. Maya is finishing a difficult day at work when a message arrives from her partner Daniel: *"Fine. I'll sort it."* Four words. No punctuation relief. No warmth. Maya has seen this pattern before — and she knows exactly what her gut says it means: dismissal. Irritation. She starts typing a response that begins *"You don't have to be like that—"*

She's used NC for three weeks. The extension intercepts the message before she finishes her reply.

Rising Action

NC surfaces a prediction overlay on WhatsApp Web: **I3 — Setting a boundary (72% confidence) / I5 — Expressing frustration (21%)** with a note: *"Daniel's messages shorten and tone formalises when he's processing stress — this pattern appears in 8 of the last 12 similar exchanges. He is most likely signalling overwhelm, not displeasure with you."*

Maya stops. She reads it twice. The drafted *"You don't have to be like that"* sits in the text box.

Climax

She deletes her draft. She types instead: *"Sounds like a rough one — I'll handle dinner. Come home when you're ready."* She sends it. Daniel replies within two minutes: *"Thank you. Really needed that."*

Resolution

The spiral didn't happen. Maya didn't need to suppress anything — she just had better information before she reacted. The conversation that would have run through the evening as a low-grade argument simply didn't start. She doesn't think of it as NC working. She thinks of it as understanding him better.

Requirements this journey reveals:

- Real-time message interception on WhatsApp Web
- Intent prediction overlay UI (non-intrusive, dismissible)
- Soul Document stress response layer driving the contextual note
- Confidence display with secondary intent visible
- Reply modification instrumentation (draft deleted = NC influenced)

Journey 2: Core Receiver — Edge Case

Persona: Priya, 41 — Co-parent, 2 years post-separation

> *"We're civil. But every message from him feels like a test I might fail."*

Opening Scene

Priya receives a message from her ex-husband Ravi about their daughter's school pickup: *"I assume you've already sorted Friday."* NC predicts: **I7 — Giving information (68% confidence)**. Priya reads it and feels immediately that this is wrong. To her, that message has never been neutral — *"I assume"* from Ravi always carries an edge of accusation. She taps the "NC got this wrong" flag.

Rising Action

NC registers the correction signal. It surfaces a follow-up prompt: *"Thanks for the correction. How would you describe what this message meant to you?"* Priya selects from a short list: **I5 — Expressing frustration.** NC notes the divergence — her Soul Document doesn't yet have enough relational history with Ravi to calibrate his patterns accurately. It displays: *"High ambiguity — NC has limited history with this sender. Treat this prediction with caution."*

Climax

Priya doesn't feel misled — she feels *seen*. NC told her it wasn't sure. It didn't pretend confidence it didn't have. She replies to Ravi factually, without escalation, and makes a mental note to trust her own read for Ravi-messages until NC has more data.

Resolution

Three weeks later, after 14 more corrected predictions on Ravi's messages, NC's accuracy on his messages reaches 81%. Priya notices she's been flagging corrections less. The calibration

happened quietly, incrementally — and she was part of it.

Requirements this journey reveals:

- "NC got this wrong" in-product correction flow
- High-ambiguity / low-confidence display state (not false confidence)
- Per-sender calibration tracking (separate from global Soul Document)
- Accuracy improvement feedback loop (correction → model signal)
- Trust recovery UX — NC acknowledging uncertainty earns more trust than masking it

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Journey 3: New User Onboarding

Persona: James, 29 — First-time user, referred by a friend

> *"My friend said it changed how he reads his girlfriend's messages. I don't really get what it does yet."*

Opening Scene

James installs the Chrome extension on a Sunday afternoon. He's mildly curious, mildly sceptical. The onboarding screen says: *"Before NC can help, it needs to know you — not your messages, just you."* It explains the Soul Document in one sentence: *"A private personality layer, stored on your device, that helps NC understand the person behind the words."*

Rising Action

Three questions. Ten minutes.

"What matters most to you — ranked, not listed?" James thinks longer than he expected to. He types: directness, then loyalty, then not wasting time. NC reflects them back as structured values and asks him to confirm or edit.

"When you're stressed or overwhelmed, how do you usually communicate?" James types: *"I go quiet. I stop texting back as fast. People think I'm angry."* NC extracts: stress response pattern — withdrawal, response latency increase, perceived as disengagement.

"Tell me about a recent conversation that was harder than it should have been." James describes a misread with his sister. NC structures a narrative entry into the Soul Document stories layer and flags it as a calibration reference.

Climax

James reviews the Soul Document summary NC has assembled. He edits one value ranking. He reads the stress response entry and thinks: *"That's exactly right."* He activates it. NC says: *"Ready. I'll start helping when messages arrive."*

Resolution

James's first interception happens three hours later — a message from his flatmate. NC's prediction feels accurate. He doesn't think about the Soul Document again. It's just working.

Requirements this journey reveals:

- 3-question guided intake UI (conversational, not form-based)
- NC-automated YAML extraction from natural language responses
- User review + edit before activation (trust gate)
- Soul Document summary view (readable, not raw YAML)
- Activation confirmation state
- Cold-start accuracy from Day 1 (no message history needed)

Journey 4: Institutional Referral

Persona: Dr. Sarah Chen, 47 — Couples therapist, private practice

> *"Half of what I do in sessions is help people decode what their partner actually meant. I'm doing it manually, one hour at a time."*

Opening Scene

Sarah hears about NC from a colleague at a conference. She's professionally cautious — she's seen apps that claim to help couples and damage them instead. She googles NC and finds the research paper. She reads the H1 validation results. She downloads the Soul Document architecture spec. She spends 40 minutes reading it.

Rising Action

What stops her from dismissing it: the privacy architecture. Local-first. No partner access. No data monetisation. She's seen enough coercive control situations to know that a relationship intelligence tool in the wrong hands is a surveillance weapon. NC has explicitly designed against this — she can see it in the spec. That earns her attention.

She clicks "Recommend to a client" — a flow that generates a shareable explainer card: what NC is, what it does, what it doesn't do, who owns the data, and a direct install link. She sends it to one couple she's been working with for six months who consistently misread each other's stress responses.

Climax

Four weeks later, the couple reports in session that they've had fewer escalation cycles. One partner says: *"I stopped assuming he was dismissing me. I started asking instead."* Sarah didn't prescribe NC — she introduced it. The couple adopted it themselves.

Resolution

Sarah recommends NC to two more clients. She emails the NC team asking about a clinical integration pathway — a read-only aggregate view for therapists, with client consent. She's not a user in the product sense. She's a referral channel and a credibility signal.

Requirements this journey reveals:

- Public-facing privacy + research summary (clinician-readable, not marketing copy)
- "Recommend to a client" referral flow with shareable explainer card
- Install link with therapist attribution (for referral tracking)
- Clinician inquiry / partnership contact path
- No requirement for therapist account or dashboard in MVP — that is Growth

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Journey 5: NC Ops / Admin

Persona: Internal NC team member — handling a quality flag

Opening Scene

A user submits a report: NC predicted I1 (Seeking Connection) on a message their partner sent during an argument. The user states the prediction was not just wrong — it felt dismissive of a genuinely threatening message pattern they had been experiencing.

Rising Action

The ops team member opens the report queue. The report includes: the message content (redacted to remove PII), the prediction made, the user's correction flag, and a free-text note. The team member reviews the pattern: this is the third report in two weeks where NC has under-predicted I5 (Expressing Frustration) in high-conflict message contexts.

This is not a one-off error — it is a systematic calibration gap in high-conflict relational contexts. The team member tags it as a model quality issue and routes it to the ML team with a priority flag.

Climax

The ML team reviews the flagged message cluster. They identify that the training data underrepresents high-conflict dyadic exchanges — the annotation dataset skewed toward moderate-valence messages. They add a targeted fine-tuning batch and flag the affected accuracy bucket in the model changelog.

Resolution

The fix ships in the next model update. The user who reported it receives a message: *"Thank you for your correction — it helped us improve NC for situations like yours."* The systematic gap

is logged in the product's known-limitations register and added to the pre-launch accuracy disclosure.

Requirements this journey reveals:

- Admin report queue with PII-redacted message + prediction log
- Correction pattern aggregation (identify systematic vs. one-off errors)
- Model quality flag + ML team routing workflow
- Model changelog and accuracy bucket tracking
- User-facing acknowledgement of correction contribution
- Known-limitations register (transparency infrastructure)

Journey Requirements Summary

Journey	Core Capabilities Revealed
1 — Receiver success	Real-time interception, intent overlay UI, Soul Document stress-pattern injection, reply modification instrumentation
2 — Receiver edge case	Correction flow, high-ambiguity display state, per-sender calibration, trust recovery UX
3 — Onboarding	Conversational intake UI, YAML extraction, user review/edit gate, Soul Document summary view, cold-start Day-1 accuracy
4 — Institutional referral	Privacy/research summary, shareable referral card, clinician inquiry path
5 — Ops/admin	Report queue, PII redaction, correction pattern aggregation, model quality routing, accuracy transparency infrastructure

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Domain-Specific Requirements

Compliance & Regulatory

GDPR — Soul Document as Special Category Data

The Soul Document encodes an individual's ranked values, decision architecture, stress responses, conflict patterns, and narrative history. This constitutes psychological profiling and maps onto GDPR Article 9 Special Category Data ("data concerning health" and data revealing personal characteristics used to profile natural persons).

Mandatory requirements from Day 1:

- Explicit, informed consent collected before Soul Document creation begins — separate from general terms of service
- Purpose limitation: Soul Document data used solely for intent prediction for the data subject; not used for model training, third-party profiling, or commercial inference without separate explicit consent
- Data minimisation: Soul Document fields limited to those with demonstrated intent prediction contribution (ablation-first principle)
- Right to erasure: Full Soul Document deletion executable by user in ≤ 3 taps; deletion confirmed to user with timestamp; no residual copy retained on NC servers
- Data Processing Agreement (DPA) required for any third-party processor that touches Soul Document data (e.g., LLM inference providers)
- GDPR Article 25 (privacy by design): local-first architecture is the compliance implementation — not a product differentiator bolted on

Sensitive communications data:

Message content intercepted by the browser extension is processed transiently and must not be stored, logged, or retained beyond the inference call:

- No message content persisted to NC servers
- No message content included in correction feedback signals without explicit per-instance user consent
- Inference pipeline ephemeral: input → prediction → output → discard

EU AI Act — Provisional Classification

NC performs automated inference on personal communications to assess communicative intent, influencing user behaviour in interpersonal relationships. This intersects with EU AI Act provisions on AI systems that make inferences about natural persons' emotional or psychological states in sensitive contexts (Article 6 + Annex III) and transparency obligations for AI systems interacting with natural persons (Article 50).

Provisional classification: Likely Limited Risk (Article 50 transparency obligations) with potential reclassification to High Risk pending regulatory guidance on emotional inference systems.

Pre-launch requirements:

- Legal opinion on EU AI Act classification obtained before any public release in EU/UK — **hard gate, not optional**
- If classified as High Risk: technical documentation (Article 11), human oversight measures (Article 14), transparency to users (Article 13), and conformity assessment required

- If classified as Limited Risk: transparency disclosure to users that intent predictions are probabilistic — not definitive
- NC's existing design (confidence display, high-ambiguity flag, correction flow) satisfies Article 50 transparency obligations in the Limited Risk scenario — document this alignment explicitly

Research Ethics — IRB Scope Boundary

- Non-study user data cannot be retroactively reclassified as research data without separate IRB amendment and individual consent
- Study participants and commercial users must be in separate data environments — no commingling of correction signals, message patterns, or behavioural telemetry
- IRB approval obtained before any study participant is recruited — **hard gate for the research track**, independent of commercial product launch timeline
- Ethics board review of safeguarding signpost design required before MVP launch

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Safeguarding Requirements

I8 Pattern Detection — Safeguarding Signpost

NC's intent label space includes I8 (Chronic self-interest): pattern-level detection of sustained disregard for the receiver's perspective across message history. This pattern is associated with coercive communication dynamics.

Signpost trigger: Automatic soft signpost when I8 pattern detected across ≥ 7 consecutive messages from a single sender with confidence ≥ 70%.

Signpost design constraints:

- Soft, non-alarming copy: "If conversations feel consistently one-sided, support is available." — not diagnostic, not alarmist
- National/regional helpline links included (UK: Refuge, Men's Advice Line; adaptable by locale)
- User can dismiss; signpost does not reappear for 30 days from dismissal
- No logging of which users triggered the signpost — signpost display is client-side only, not reported to NC servers
- Signpost copy reviewed by domestic abuse / coercive control specialist before launch — **hard gate**

Domestic abuse scenario — architectural prohibitions (v1 and permanent):

- Partner access to another user's Soul Document by any mechanism — prohibited
- Shared device detection that would expose Soul Document to a co-user — prohibited
- Any feature usable as a surveillance instrument by an abusive partner — prohibited

- These prohibitions must be included in the engineering threat model and reviewed at every major feature addition

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Technical Constraints

Browser extension — Chrome MV3 host permissions:

- Request only minimum host permissions required for supported channels (Gmail, WhatsApp Web, Slack web, iMessage web)
- Plain-language explanation of what is accessed and why, at permission request time
- No speculative permissions for unsupported channels
- Comply with Chrome Web Store single-purpose policy
- Chrome Web Store pre-review submission required before launch — **hard gate**

Mobile app — platform permissions:

- iOS: Soul Document encrypted at rest via iOS Keychain; biometric/PIN gate via LocalAuthentication framework
- Android: Soul Document encrypted at rest via Android Keystore; biometric/PIN gate on API 26+ (Android 8.0+); minimum supported version API 26, covering ~95% of active Android devices
- Neither platform: real-time message interception in MVP — Growth feature requiring independent platform compliance review before shipping

Apple App Store — iOS mental health guidelines (Guideline 1.4.2):

- App must not claim diagnostic or therapeutic capabilities
- Wellbeing claims must be accurate and not misleading
- Crisis resource (safeguarding signpost) required
- Legal review of App Store submission copy before submission — **hard gate**

Google Play Store — mental health / wellness app policy:

- Equivalent requirements to Apple: no diagnostic claims, accurate wellbeing claims, crisis resource included
- Sensitive permissions (notification access, Accessibility Services) not required in MVP — deferred to Growth mobile interception feature
- Legal review of Play Store submission copy before submission — **hard gate**

Integration Requirements

LLM inference providers:

Any LLM provider processing Soul Document content at inference time is a GDPR data processor and requires:

- Signed Data Processing Agreement (DPA) with zero data retention clause
- Confirmation that inference inputs are not retained for model training
- EU/UK data residency or Standard Contractual Clauses (SCCs) if processing occurs outside UK/EEA
- Preferred providers: those with explicit zero-retention inference modes (e.g., OpenAI Enterprise, Anthropic API with DPA)

Annotation pipeline separation:

The 2,000-message annotated dataset used for model training is governed by separate consent from message contributors and must not be linked to any commercial product user profile. Separate data environments enforced architecturally.

Risk Register — Domain-Specific

Risk	Likelihood	Impact	Mitigation
EU AI Act reclassified as High Risk post-launch	Medium	High — conformity assessment required, potential launch pause	Legal opinion pre-launch; architecture built to High Risk standard where feasible
Chrome Web Store rejects extension for host permissions	Medium	High — blocks primary distribution channel	Pre-review submission; single-purpose justification; legal review of privacy policy
Google Play rejects app for sensitive permission claims	Low	Medium — blocks Android distribution	MVP avoids sensitive permissions entirely; legal review of submission copy
Soul Document accessed by abusive partner via shared device	Low	Critical — user safety	Biometric/PIN gate; domestic abuse scenario in engineering threat model
IRB data and commercial data commingled	Low	High — ethics violation, research invalidation	Separate data environments; architectural enforcement

I8 signpost triggers false positive — user distressed	Medium	Medium — user harm, trust damage	Specialist copy review; high confidence threshold (≥70%); 30-day suppression after dismiss
LLM provider retains Soul Document for model training	Low	High — GDPR breach	DPA with zero-retention clause; audit rights; preferred zero-retention providers

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Innovation & Novel Patterns

Detected Innovation Areas

1. Person-Aware Intent Modelling via Soul Document

Every existing NLP-based communication tool — Grammarly, Crystal Knows, Replika, tone analysers — operates on the message in isolation. NC is the first consumer product to condition intent prediction on a structured model of *who the person is*: their ranked values, stress response patterns, conflict archetypes, and relational narrative history.

The formal model: **P(intent | message, soul_document, relational_context)** — where the Soul Document is a persistent, user-owned personality layer injected at inference time. This is architecturally distinct from:

- Fine-tuning (Soul Document is not trained into the model — it is a prompt-layer injection, preserving updateability and user control)
- RAG from interaction history (Soul Document captures *who the person is*, not what they said last week)
- Persona APIs (Crystal Knows infers personality from public data; NC captures self-reported, private, nuanced personality at much higher resolution)

The commercial differentiator: NC knows who they are, not just what they said. This makes every prediction contextually grounded in the sender's actual psychology — not a generic interpretation of the words.

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2. Intent Gap as a Formal Metric

NC introduces a novel measurable quantity to consumer communication: the **intent gap** — the divergence between what was meant (sender intent, SI) and what was received (perceived intent, PI), formalised as:

Intent Gap = JS-divergence(P(SI), P(PI))

Where SI is captured via sender self-annotation (gold standard) and PI is NC's real-time prediction of how the receiver will interpret the message. This is not a metaphor — it is a computable, trackable quantity that has never been operationalised at consumer scale.

The scientific significance: this makes relational misalignment *measurable* — allowing NC to track whether intent gaps narrow over time for users who train with the Soul Document, providing the mechanistic evidence for H1 (intent prediction accuracy), H2 (conflict de-escalation), and H3 (relational satisfaction).

The product significance: users can see, concretely, that NC improved their understanding of a specific sender. Improvement is not a feeling — it is a tracked delta.

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3. Receiver-Side Biological Window Targeting

The intervention timing in NC is grounded in a specific neurobiological mechanism: the 40–800ms window between message receipt and amygdala threat-tagging. Within this window, the brain is still in interpretive mode — threat has not yet been embedded. NC's interception goal is to surface the intent prediction *before* the amygdala fires, inserting a re-appraisal signal into the cognitive pathway before the emotional response is complete.

This is categorically different from post-hoc communication coaching (Reflectly, therapy apps) or pre-send nudges (Grammarly tone detection). NC intervenes at the *moment of reception*, in the *biological window where intervention has maximum effect*. The < 800ms latency target is not a UX performance target — it is a neuroscience-derived constraint.

No competitor has articulated, let alone designed to, this specific intervention window.

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4. Cold-Start Accuracy via Personality Injection

The standard cold-start problem in AI personalisation: the system is useless until it has accumulated enough interaction history to calibrate. NC solves cold-start differently — the Soul Document provides a rich personality layer from Day 1, before any message history exists. A user who completes the Soul Document intake on onboarding has a higher-accuracy prediction engine from their *first* message than any interaction-history-based system achieves after months of use.

This creates a compounding advantage: the Soul Document gets richer over time, meaning NC's accuracy trajectory is fundamentally different from competitor approaches (steep initial improvement vs. NC's high baseline with continued refinement).

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5. Hypothesis-Driven Product Development — The Product Is the Experiment

NC is being built as a scientific instrument that happens to be a consumer product. The three core hypotheses (H1: intent prediction accuracy, H2: conflict reduction, H3: relational satisfaction) are pre-registered, falsifiable, and designed to produce a publishable result regardless of outcome. If NC fails to improve relational outcomes, that is a contribution to the literature.

NC's RCT design means the product team will have causal evidence — not correlational — for NC's effect on communication outcomes within the fellowship year. This evidence is the foundation for institutional credibility, clinical adoption, and research partnership that no competitor can acquire through product iteration alone.

Market Context & Competitive Landscape

The white space NC occupies:

Category	Example products	What they do	What they miss
Pre-send tone detection	Grammarly, Hemingway	Analyse your draft before you send	Do not address received message interpretation; no soul model; no relational history
Personality profiling	Crystal Knows, DISC tools	Infer personality from public data	No real-time message interception; no intent prediction; professional context only
Post-hoc relationship coaching	Gottman apps, Lasting	Reflect on past interactions	No real-time intervention; no message-level analysis; therapeutic framing
General AI assistants	ChatGPT, Claude	Answer questions including "what did they mean?"	No soul document; no persistent relational context; user-initiated only; no scientific validation
Mindfulness / emotional regulation	Headspace, Calm	Regulate your own state	Do not address the source of the trigger; no message analysis

NC's unoccupied position: Real-time, receiver-side, person-aware intent prediction grounded in neuroscience and validated by a pre-registered RCT. No product in any of these categories is

building toward this position.

Closest emerging threat: Large AI platforms adding relationship/communication intelligence to general assistants. NC's moat: the Soul Document architecture (depth of person-model), the relational dataset (annotated intent-gap pairs), and the scientific credibility (RCT results). These are not replicable by prompt engineering.

Validation Approach

Innovation	Validation Method
Person-aware modelling	Ablation study within RCT — accuracy with Soul Document active vs. disabled on matched message sets
Intent gap metric	Pre/post JS-divergence measurement in RCT — reduction in intent gap for NC users vs. control group
Biological window targeting	Instrumented latency measurement — sub-800ms confirmed at 95th percentile of production traffic before launch
Cold-start via personality injection	Day 1 accuracy (Soul Document only) vs. interaction-history-based system at Day 30 — target: NC Day 1 ≥ competitor Day 30
Hypothesis-driven development	RCT pre-registered before recruitment; results published regardless of direction

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Risk Mitigation

Innovation Risk	Likelihood	Mitigation
Soul Document accuracy contribution smaller than expected	Medium	Modular architecture — additional intake dimensions added iteratively without model retraining
Intent gap metric too noisy to be meaningful	Medium	Pre-launch calibration on annotation dataset; JS-divergence threshold tuned before RCT
< 800ms latency not achievable with full Soul Document injection	Medium	Tiered inference: lightweight prediction first (< 200ms), Soul Document enrichment layer second (< 600ms additional)

Large platform replicates person-aware intent prediction	Low (12-month horizon)	Moat is relational dataset + Soul Document depth + RCT credibility; file provisional patents on Soul Document architecture + intent gap metric before launch
RCT results null or negative	Low-Medium	Pre-registered null result is publishable; NC pivots on evidence; Soul Document architecture retains independent value regardless of RCT outcome
Soul Document becomes a coercive control tool	Low	Architectural prohibition on partner access; domestic abuse threat model reviewed at every feature addition

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Multi-Surface Specific Requirements

Project-Type Overview

NC is a **dual-surface product** with asymmetric responsibilities across surfaces:

Surface	Primary Role	Runtime Context
Chrome Extension (MV3)	Real-time intent prediction — the core value delivery layer	Desktop browser, persistent content script, < 800ms inference loop
Mobile App (React Native)	Soul Document management + async reflection — the depth and trust layer	iOS + Android, offline-first, notification-driven

The surfaces are architecturally coupled at the Soul Document layer (shared encrypted data model) but independently deployable. The extension can function without the mobile app; the mobile app without the extension provides Soul Document management only. Desktop is the primary interception channel in MVP; mobile is the onboarding and retention layer.

Platform Requirements

Chrome Extension — MV3

Requirement	Specification

Manifest version	MV3 — no legacy MV2 fallback
Background execution	Service worker (not persistent background page) — all persistent state managed via `chrome.storage`
Content script injection	Declarative injection via `content_scripts` manifest key; dynamic injection via `chrome.scripting.executeScript` for late-loading SPAs
Host permissions	Declared in manifest: `mail.google.com`, `web.whatsapp.com`, `app.slack.com`, `messages.google.com` — no wildcard permissions
Storage	`chrome.storage.local` for extension state; Soul Document cloud sync deferred to Growth
Minimum Chrome version	Chrome 102+ (MV3 stable)
Cross-browser	Chrome only in MVP; Firefox and Safari Web Extensions as Growth — separate compliance review per browser

Mobile App — React Native

Requirement	Specification
Framework	React Native (latest stable LTS at build time)
iOS minimum	iOS 15.0+ (~95% of active iOS devices)
Android minimum	API 26 / Android 8.0 (~95% of active Android devices)
Navigation	React Navigation v6 (stack + tab navigator)
State management	Zustand (lightweight; no Redux overhead for MVP scope)
Offline-first	Full offline capability required — all Soul Document read/write operations must function without network
Build tooling	Expo managed workflow for MVP; eject to bare workflow if native modules require it at Growth

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Device Permissions

Chrome Extension permissions matrix:

Permission	Purpose	MVP	Store Justification
`host_permissions` (4 domains)	Content script injection for message interception	Yes	Single-purpose: intent prediction on declared messaging platforms
`storage`	Extension state persistence	Yes	Standard — no user data stored server-side
`scripting`	Dynamic content script execution	Yes	Required for SPA platforms (Slack, WhatsApp Web) that load asynchronously
`activeTab`	Overlay injection into active tab	Yes	Scoped to active tab only — not persistent access
`notifications`	System notification for high-ambiguity alert	No	Deferred — overlay-based MVP does not require

Mobile App permissions matrix:

Permission	Platform	Purpose	MVP
`FaceID` / `TouchID`	iOS	Biometric gate on Soul Document	Yes
`USE_BIOMETRIC` / `USE_FINGERPRINT`	Android	Biometric gate on Soul Document	Yes
`POST_NOTIFICATIONS` (Android 13+)	Android	Async reflection push prompts	Yes
`UserNotifications`	iOS	Async reflection push prompts	Yes
Camera / Location / Microphone	Both	Not required	No — never in MVP
`RECEIVE_SMS` / Accessibility Services	Android	Mobile real-time interception	No — Growth only

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Offline Mode

Extension: Not applicable — inherently online. If inference API unreachable: overlay suppressed, no error state shown, message reading unblocked. NC never delays the user.

Mobile App — fully offline for all MVP features:

Feature	Offline behaviour
Soul Document read / view	Fully offline — local storage only
Soul Document edit	Fully offline — writes to encrypted local store; sync queued when online
Onboarding intake	Fully offline — YAML extraction runs locally; falls back to manual intake if LLM inference unavailable
Async reflection prompts	Scheduled locally — local notification delivery; no push server dependency in MVP
Reflection journal entries	Fully offline — stored locally; no server upload without explicit user action

Data layer: `expo-sqlite` for structured Soul Document storage + `react-native-keychain` for AES-256 encryption key management. SQLite database file encrypted at rest on both platforms via Android Keystore (Android) and iOS Keychain (iOS).

Push Notification Strategy

MVP: Local notifications only — no push server infrastructure required.

Notification type	Trigger	Delivery	Phase
Daily reflection prompt	User-configured time (default: 8pm)	Local	MVP
Weekly pattern summary	Sunday 10am	Local	MVP
High-ambiguity alert (I8 threshold)	Client-side pattern detection	Local	MVP
Accuracy improvement confirmation	Post-correction feedback	Local	MVP

Consent network request	Partner requests mutual intent sharing	Push (APNs / FCM)	Growth

Push infrastructure (Growth): APNs + FCM. Zero PII in push payload — notification body assembled client-side from payload token.

Technical Architecture Considerations

Content script ↔ Service worker communication:

MV3 service workers are ephemeral (spin down after ~30s inactivity). All persistent state must be in `chrome.storage.local`. Content scripts communicate via `chrome.runtime.sendMessage` / `chrome.runtime.onMessage`. If the service worker is inactive at message receipt, Chrome restarts it — content script handles startup latency with retry (max 2 retries, exponential backoff) before silent degradation.

Inference pipeline:

```
Content script (message receipt)
  → extract message text + sender ID
  → chrome.runtime.sendMessage → service worker
    → retrieve Soul Document context
(chrome.storage.local)
    → construct inference payload
    → POST to NC inference API (target: < 600ms)
    → receive intent prediction
  → chrome.runtime.sendMessage → content script
    → render intent overlay UI (target: < 200ms)
Total end-to-end: < 800ms
```

Tiered inference fallback (if Soul Document injection exceeds latency budget):

1. Lightweight prediction (message only) — < 200ms — display immediately
2. Soul Document enrichment pass — < 600ms additional — update overlay if enriched prediction differs significantly

React Native ↔ Extension data bridge (MVP):

No live data bridge in MVP. Soul Document authored on mobile; extension reads a cached copy synced via QR-code-based device pairing or manual export (iCloud / Google Drive). Cloud sync with E2E encryption is a Growth feature.

Accessibility — WCAG 2.1 AA

The intent prediction overlay is NC's primary UI element. Non-negotiable accessibility requirements:

- All intent labels have accessible text alternatives — no icon-only display
- Overlay is keyboard navigable (Tab / Enter / Escape)
- Confidence display does not rely solely on colour (colour-blind users)
- Overlay respects `prefers-reduced-motion` — no animation if reduced motion set
- React Native: all interactive elements have `accessibilityLabel` props; screen reader tested on VoiceOver (iOS) and TalkBack (Android)

Store Compliance Checklist

(Full requirements in Domain-Specific Requirements — this is the implementation checklist)

Check	Chrome Web Store	App Store	Play Store
Single-purpose declaration	✓ Required	N/A	N/A
Privacy policy URL	✓ Required	✓ Required	✓ Required
Host permissions justification	✓ Each domain justified	N/A	N/A
No diagnostic / therapeutic claims	N/A	✓ Guideline 1.4.2	✓ Wellness policy
Crisis resource present	N/A	✓ Required	✓ Required
Legal review of submission copy	✓ Hard gate	✓ Hard gate	✓ Hard gate
Pre-review submission	✓ Hard gate	N/A	N/A
Data safety / privacy nutrition label	N/A	✓ Required	✓ Required

Data safety declaration (both mobile stores): No user data transmitted to NC servers in MVP. LLM inference payload (message text + Soul Document excerpt) declared as "App functionality — encrypted in transit — not sold." Full data deletion supported (≤ 3 taps, timestamp confirmation).

Implementation Sequencing

Extension first, mobile second. The extension is the value-delivery surface. Ship extension in working state before mobile app is available. The extension must include a fallback inline onboarding flow (simplified Soul Document intake) for the pre-mobile-app period — users cannot be blocked from using NC because the mobile app is not yet released.

Project Scoping & Phased Development

MVP Strategy & Philosophy

MVP Approach: Platform MVP

NC's MVP is not a feature MVP (a single polished feature) or a revenue MVP (a monetisable product). It is a **platform MVP** — the goal is to ship the infrastructure that makes all future NC applications possible: the Soul Document layer, the intent prediction engine, and the real-time interception surface. Every scoping decision protects this platform layer and defers application-layer features.

The test of a valid MVP feature: *"Does this teach us something we cannot learn any other way, or does it protect a user from harm?"* If neither, it is Growth.

MVP Philosophy: One channel, one use case, done properly.

Ship WhatsApp Web intent prediction with Soul Document v1 on Chrome. Prove the core loop: message arrives → NC intercepts → intent predicted → user modifies response. Everything else follows from this proof.

MVP Team Requirements:

Role	Responsibility	FTE
ML / AI engineer	Intent prediction model, Soul Document injection pipeline, inference API	1
Frontend / Extension engineer	Chrome MV3 extension, content script, overlay UI	1

React Native engineer	Mobile app (Soul Document intake, reflection layer, cross-platform)	1
Product / Researcher	Soul Document design, annotation pipeline, RCT protocol, IRB liaison	1 (founder)
Legal / compliance	GDPR DPA, EU AI Act review, store submission copy	0.25 (retained advisor)

Minimum viable team: 3 engineers + 1 founder-researcher.

MVP Feature Set — Phase 1

Core journeys supported in MVP:

- ✅ Journey 1: Core Receiver — Success Path (primary)
- ✅ Journey 2: Core Receiver — Edge Case (trust recovery)
- ✅ Journey 3: New User Onboarding (simplified inline + full mobile)
- ✅ Journey 4: Institutional Referral (lightweight — referral card only, no therapist account)
- ⚠️ Journey 5: Ops / Admin (manual process — no admin UI)

Must-Have Capabilities:

Capability	Surface	Rationale
WhatsApp Web real-time message interception	Extension	Primary channel; highest intimate-relationship traffic
Intent prediction overlay UI (non-intrusive, dismissible)	Extension	Core value delivery
8-category intent label space with confidence display	Extension	Validated in annotation study; satisfies EU AI Act Article 50
High-ambiguity / low-confidence flag	Extension	Trust recovery UX
"NC got this wrong" correction flow	Extension	Per-sender calibration; accuracy feedback loop
Tiered inference (lightweight first, Soul Document enrichment second)	Extension + API	Latency budget compliance; graceful degradation

Simplified 3-question inline onboarding (extension fallback)	Extension	Unblocks extension launch before mobile app ships
Soul Document v1 — full guided intake (conversational, 3 questions)	Mobile	Rich cold-start accuracy; biometric-gated trust
Soul Document summary view + edit	Mobile	User review and control of personality layer
Soul Document YAML extraction from natural language	Mobile	NC-automated extraction; user answers in prose
AES-256 encrypted local storage (Keychain / Keystore)	Both	Privacy-by-design hard constraint
Biometric / PIN gate on Soul Document	Mobile	Domestic abuse scenario protection — hard constraint
Full data deletion (≤ 3 taps + timestamp confirmation)	Both	GDPR right to erasure — hard constraint
Daily reflection prompt (local notification)	Mobile	Retention mechanism; neuroplasticity window engagement
Reply modification instrumentation (passive)	Extension	Core success metric — no prompts, passive tracking only
I8 safeguarding signpost (client-side, no server logging)	Extension	User safety — hard constraint
QR-code device pairing (extension ↔ mobile Soul Document sync)	Both	MVP data bridge — no cloud infrastructure required
Shareable institutional referral card	Extension / Web	Journey 4 — therapist attribution link
PII-redacted correction logs (structured for manual review)	API	Ops / admin captured correctly even if surfaced manually
GDPR consent gate (pre-Soul Document creation)	Both	Hard gate — explicit consent separate from ToS

MVP explicitly excludes:

Feature	Phase
Gmail, Slack, iMessage Web interception	Growth
Firefox / Safari extensions	Growth
Mobile real-time message interception (iOS or Android)	Growth
Admin UI / report queue dashboard	Growth
Cloud sync / multi-device Soul Document	Growth
Soul Document v2 (Layers 4–6)	Growth
Revenue / payment infrastructure	Growth
Partner consent network	Vision
Clinician dashboard	Growth
Voice channel support	Vision
Cross-language support	Growth

Post-MVP Features — Phase 2 (Growth)

Target: Months 7–18 post-MVP launch

Channel expansion: Gmail, Slack web, iMessage web (Safari Web Extension, separate review pathway)

Mobile interception: iOS Share Extension (WhatsApp); Android notification listener (WhatsApp + SMS, pending Play Store policy review). Both require independent compliance review and domestic abuse threat model re-review before shipping.

Soul Document deepening: v2 intake (Layers 4–6: Narrative History, Relational Maps, Temporal Patterns); automated story tagging; per-sender calibration history surfaced to user

Platform enhancements: Cloud sync with E2E encryption; multi-device access; in-app accuracy trajectory view; weekly pattern insight notifications

Ops tooling: Admin report queue; correction pattern aggregation; model quality routing; known-limitations register (public)

Institutional pathway: Therapist account (read-only, consent-gated); clinician privacy + research summary; partnership inquiry → LOI workflow

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Post-MVP Features — Phase 3 (Vision)

Target: 18–36 months, post-RCT results

- **Consent network:** Bilateral opt-in intent sharing; ethics review + coercive control architecture review required before build
- **Neuroplasticity programme:** Structured 60-day programme grounded in H3 mechanism
- **Voice channel:** Real-time intent prediction on voice messages and calls; new model architecture required
- **Clinician dashboard:** Aggregate pattern trends (not message content); consent-gated
- **Cross-language support:** Spanish, Mandarin, Hindi
- **Foundation model open release:** Intent prediction model + annotated dataset as open-source scientific infrastructure
- **Institutional integration API:** Headless NC engine for therapy platforms, HR tools, co-parenting apps
- **Temporal drift detection:** NC detects divergence between a user's Soul Document and a sender's evolving communication patterns — a leading indicator of relational deterioration before either party is consciously aware of it
- **Neurodivergent calibration programme:** ND-specific Soul Document intake and intent model calibration, validated by a three-group RCT (ND vs. neurotypical receiver; ND vs. NT sender); addresses the intent-expression gap unique to ADHD and ASD communicators

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Risk Mitigation Strategy

Technical risks:

Risk	Mitigation
WhatsApp Web DOM changes break content script	Versioned content script adapters; automated DOM regression tests on CI
< 800ms latency not achievable at launch	Tiered inference architecture — acceptable degradation path defined
Soul Document YAML extraction insufficient for cold-start accuracy	Pre-launch calibration on extraction prompts; user review/edit gate catches errors
MV3 service worker drops messages	Stateless inference pipeline — each message a self-contained request; no in-memory state required

Market risks:

Risk	Mitigation
WhatsApp Web blocks extension	Modular channel adapter architecture — Gmail is backup primary channel
User adoption slower than expected	Waitlist pre-launch; therapist referral channel seeded before general availability; RCT participants as early adopter cohort
Trust barrier to Soul Document completion	Inline 3-question fallback means users experience NC value before committing to full Soul Document intake
Large platform copies core mechanic	Patent filing on Soul Document architecture + intent gap metric before launch

Resource risks:

Risk	Mitigation
ML engineer unavailable at launch	Pre-trained base model via API (Anthropic / OpenAI); NC adds Soul Document injection — not training from scratch in MVP
React Native scope creeps	Mobile MVP strictly Soul Document management + reflection — no feature additions without explicit scope decision
Legal / compliance delays store submission	Store review initiated 6 weeks before target launch; legal review begins at feature-freeze, not at submission
Fellowship year ends before 300-user target	Month 9 checkpoint: if active users < 150, trigger therapist referral acceleration and institutional partnership push

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MVP Launch Gate Sequence

Milestone	Gate	Timing
Inference API live — 75% accuracy on held-out test set	Accuracy validation	Pre-alpha
Extension inline onboarding + overlay UI complete	WCAG 2.1 AA audit; < 800ms at P95	Pre-alpha

Mobile Soul Document intake + QR sync complete	Biometric gate tested on iOS + Android	Alpha
IRB approval for RCT	Hard gate — no study recruitment before this	Alpha
GDPR consent flow + data deletion verified	Legal sign-off	Beta
EU AI Act legal opinion received	Hard gate — no public launch before this	Beta
I8 safeguarding signpost reviewed by specialist	Hard gate	Beta
Chrome Web Store pre-review submitted and accepted	Hard gate	Beta
App Store + Play Store legal review + submission accepted	Hard gate	Launch

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Functional Requirements

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1. Message Interception & Intent Prediction

- **FR01:** The system can detect when a new message arrives on a supported messaging platform in an active browser tab
- **FR02:** The system can extract message text and sender identity from a supported messaging platform without storing message content beyond the inference call
- **FR03:** The system can classify a received message into one of eight intent categories with an associated confidence score. The intent label space must cover social nuance categories including irony and sarcasm, passive aggression, and deference and politeness — these are not edge cases but high-frequency sources of the intent gap in intimate communication
- **FR04:** The system can identify when a message has high entropy across intent categories and flag it as high-ambiguity
- **FR05:** The system can apply Soul Document context to enrich a base intent prediction with person-specific calibration
- **FR06:** The system can detect when the inference API is unreachable and degrade gracefully without blocking or delaying message reading

- **FR07:** The system can perform tiered inference — producing a lightweight prediction first, then an enriched prediction when Soul Document context is available

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2. Intent Prediction Overlay UI

- **FR08:** A receiver can view an intent prediction overlay when a new message arrives, before composing a reply
- **FR09:** A receiver can view the primary predicted intent label and confidence score in the overlay
- **FR10:** A receiver can view a secondary intent label and confidence score when the prediction distribution is non-uniform
- **FR11:** A receiver can view a contextual note explaining why NC predicts this intent based on sender patterns from the Soul Document
- **FR12:** A receiver can dismiss the intent prediction overlay without taking any action
- **FR13:** A receiver can flag a prediction as incorrect directly from the overlay
- **FR14:** A receiver can view a high-ambiguity indicator when NC cannot predict intent with sufficient confidence
- **FR15:** A receiver can view the I8 safeguarding signpost when a sustained coercive communication pattern is detected from a sender

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3. Soul Document Management

- **FR16:** A user can create a Soul Document by completing a guided conversational intake
- **FR17:** The system can extract structured personality data (values, stress responses, narrative story) from a user's natural language intake responses
- **FR18:** A user can review the structured Soul Document summary before activating it
- **FR19:** A user can edit any field in their Soul Document summary before or after activation
- **FR20:** A user can activate their Soul Document to enable person-aware intent prediction
- **FR21:** A user can view their complete Soul Document in a readable summary format at any time
- **FR22:** A user can complete a simplified 3-question intake via the browser extension when the mobile app is not available
- **FR23:** A user can access their Soul Document only after passing biometric or PIN authentication
- **FR24:** A user can permanently delete all NC data including their Soul Document in three or fewer steps, with deletion confirmed by timestamp

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4. Correction & Calibration

- **FR25:** A receiver can submit a correction when NC's intent prediction does not match their interpretation
- **FR26:** A receiver can select the correct intent label from the 8-category label space when submitting a correction
- **FR27:** The system can record corrections as a per-sender calibration signal linked to the receiver's Soul Document context
- **FR28:** The system can surface an acknowledgement to a user when their correction has contributed to a model improvement
- **FR29:** The system can track correction patterns across multiple messages from the same sender to detect systematic calibration gaps

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5. Onboarding & User Activation

- **FR30:** A new user can install the browser extension and reach a working intent prediction state without requiring the mobile app to be installed
- **FR31:** A new user can understand what NC does, what the Soul Document is, and what data NC accesses, before providing any personal information
- **FR32:** A new user can complete the GDPR consent process as a distinct step before Soul Document creation begins
- **FR33:** The system can present the Soul Document onboarding as a conversational flow, not a form
- **FR34:** A user can pair their browser extension with their mobile app via QR code to sync their Soul Document across surfaces

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6. Reflection & Retention

- **FR35:** A user can receive a daily async reflection prompt at a user-configured time via local notification
- **FR36:** A user can receive a weekly pattern summary notification highlighting recurring communication themes
- **FR37:** A user can write and save a reflection journal entry in response to a prompt
- **FR38:** A user can view their reflection journal history within the mobile app

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7. Institutional Referral

- **FR39:** A clinician or institutional user can generate a shareable referral card for NC containing product description, privacy architecture summary, and a direct install link
- **FR40:** The system can attribute installs to a referring clinician via a unique referral link for tracking purposes
- **FR41:** A clinician or institutional user can submit an inquiry about partnership or clinical integration via a contact pathway

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8. Privacy, Safety & Compliance

- **FR42:** The system can store the Soul Document on-device in encrypted form without transmitting it to NC servers except as part of a transient inference payload
- **FR43:** The system can process message content transiently — input to inference, prediction returned, message content discarded — with no server-side persistence of message text
- **FR44:** The system can display the I8 safeguarding signpost with regional crisis resource links when a coercive communication pattern threshold is met, without logging which users triggered it
- **FR45:** The system can suppress the I8 safeguarding signpost for 30 days after a user dismisses it
- **FR46:** The system can enforce architectural prohibition on any mechanism that allows one user to access another user's Soul Document
- **FR47:** A user can view a plain-language explanation of what data NC accesses, stores, and transmits at any point during onboarding or post-activation

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9. Ops & Quality (Internal)

- **FR48:** The system can capture correction events as PII-redacted structured logs accessible to the NC operations team
- **FR49:** An NC operations team member can review correction reports containing redacted message content, the prediction made, and the user's correction
- **FR50:** The system can route a quality flag to the ML team when a correction pattern indicates a systematic model calibration gap

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Capability Contract — Coverage Validation

MVP Capability	FR Coverage
WhatsApp Web interception	FR01, FR02, FR06
Intent prediction overlay	FR08–FR15
8-category intent + confidence	FR03, FR09, FR10
High-ambiguity flag	FR04, FR14
Correction flow	FR25–FR29
Tiered inference	FR07
Inline onboarding (extension)	FR22, FR30, FR31

Soul Document v1 intake (mobile)	FR16–FR21
Biometric / PIN gate	FR23
Full data deletion	FR24
Daily reflection prompt	FR35
I8 safeguarding signpost	FR15, FR44, FR45
QR pairing	FR34
Institutional referral card	FR39, FR40
PII-redacted correction logs	FR48, FR49
GDPR consent gate	FR32
Architectural prohibition (partner access)	FR46

> ⚠️ ***Capability Contract is now binding.*** *50 FRs across 9 capability areas. Any feature not listed does not exist in the final product unless explicitly added via a scope change decision.*

Non-Functional Requirements

Product Experience Philosophy

NC's interaction register is **warm, quiet, and non-intrusive**. Every design decision — overlay placement, copy tone, prediction confidence display, correction flow — must be consistent with the following principles, which UX designers must treat as binding:

- **NC is a knowing friend, not a clinical tool.** It surfaces insight without judgment. It never implies the user is wrong to feel what they feel — only that there may be more to the story.
- **NC earns trust by admitting uncertainty.** A high-ambiguity flag is not a failure state — it is the product behaving with integrity. Designs that hide uncertainty erode trust faster than designs that surface it.
- **NC never interrupts.** The overlay is available — never intrusive. It appears only on incoming messages, never during composition. It can be dismissed in one action. It never blocks.
- **NC is consent-first, transparency-by-design.** The user always knows what NC can see, what it stores, and how to remove it. Privacy is not fine print — it is the product's primary trust signal.

- **NC's language is human-scale, not algorithmic.** Prediction labels and contextual notes must be written in the language a thoughtful friend would use, not the language of a classification system.

These principles are not aspirational — they are testable. UX review at each sprint should include an explicit check: does this design feel like a knowing friend or a surveillance tool?

Performance

- **NFR-P1:** The intent prediction overlay must be visible within **800ms** of message receipt at P95 of production traffic. Measured: message DOM mutation event → overlay render complete. This is a neuroscience-derived hard constraint — beyond 800ms the amygdala threat-tagging response is complete.
- **NFR-P2:** Lightweight prediction (message only): rendered within **200ms** at P95. Soul Document enrichment layer: delivered within **600ms** additional (total ≤ 800ms) at P95. If enrichment exceeds budget: lightweight prediction displayed; enriched update suppressed.
- **NFR-P3:** Browser extension must reach operational state within **1 second** of browser startup on Chrome 102+.
- **NFR-P4:** React Native app must display first interactive screen within **2 seconds** of launch on minimum supported device (mid-range Android API 26; mid-range iPhone iOS 15).
- **NFR-P5:** All Soul Document read, write, and edit operations must complete within **500ms** including AES-256 encryption/decryption on the mobile app.
- **NFR-P6:** All mobile features designated as offline-capable (FR16–FR24, FR35–FR38) must maintain identical performance characteristics online and offline — no network round-trip introduced into offline flows.

Security & Privacy

- **NFR-S1:** Soul Document database file encrypted at rest with AES-256. Encryption key stored exclusively in platform secure storage (iOS Keychain / Android Keystore). Key never written to disk unencrypted, logged, or transmitted to NC servers. Verified by security audit before launch.
- **NFR-S2:** Message text extracted by the content script must not persist beyond the inference call lifecycle. No message content written to `chrome.storage`, logged server-side, or retained in any memory structure after prediction is returned.
- **NFR-S3:** All inference payloads transmitted over HTTPS with TLS 1.3 minimum. Certificate pinning applied to the NC inference API endpoint in the extension.
- **NFR-S4:** No API endpoint, data structure, or user flow may expose one user's Soul Document to another user's session. Verified by threat model review at each major feature addition. Architectural enforcement — not policy-only.

- **NFR-S5:** Biometric / PIN gate on Soul Document uses platform-native authentication APIs exclusively (iOS `LocalAuthentication`, Android `BiometricPrompt`). Gate re-challenges after app is backgrounded for > 60 seconds.
- **NFR-S6:** Soul Document fields limited to those with demonstrated intent prediction contribution via ablation study. Fields with no measurable accuracy contribution must be removed. New fields added only after ablation validation confirms prediction contribution.
- **NFR-S7:** User-initiated deletion removes all NC data (Soul Document, correction history, reflection journal, per-sender calibration, extension local storage) within **10 seconds**, confirmed to user with timestamp. No residual data retained on NC servers.
- **NFR-S8:** The I8 safeguarding signpost trigger event must not be logged to NC servers or linked to any user identifier. Detection is client-side only.
- **NFR-S9:** LLM inference provider must operate under a signed DPA with explicit zero data retention for inference inputs, including audit rights. Violation is a GDPR breach and a product launch blocker.

Reliability & Resilience

- **NFR-R1:** NC must never prevent a user from reading or responding to messages. Any NC component failure (inference API timeout, service worker inactive, overlay render error) results in NC becoming invisible — not blocking.
- **NFR-R2:** NC inference API must maintain **99.5% uptime** during active hours (06:00–24:00 UK time) once the product reaches 100+ daily active users.
- **NFR-R3:** Each inference request must be stateless — fully self-contained with all required context in the request payload. Service worker restart mid-flow must not drop the prediction.
- **NFR-R4:** Complete network loss must not cause data loss for any Soul Document write operation. Writes committed to encrypted local database before any network sync attempt.
- **NFR-R5:** Chrome extension updates must not interrupt an active inference session. Content script must gracefully restart without displaying an error to the user.

Accessibility

- **NFR-A1:** The intent prediction overlay (extension) and all mobile app screens must conform to **WCAG 2.1 Level AA**. Required under UK Equality Act 2010 for public-facing digital services.
- **NFR-A2:** Confidence display and intent label indicators must communicate meaning via at least two distinct visual channels (e.g., colour + icon, colour + text label). No information conveyed by colour alone.

- **NFR-A3:** Intent prediction overlay must be fully operable via keyboard. Tab order: overlay focus → primary intent label → secondary label → dismiss → flag incorrect. No pointer required.
- **NFR-A4:** All overlay elements must have accessible text alternatives compatible with NVDA/JAWS (Windows) and VoiceOver (macOS). React Native: all interactive elements have `accessibilityLabel` and `accessibilityRole` props. Screen reader compatibility tested on VoiceOver (iOS) and TalkBack (Android) before launch.
- **NFR-A5:** Overlay appearance and animated elements must respect `prefers-reduced-motion`. If set: overlays appear/disappear instantly — no animation or transition.
- **NFR-A6:** Mobile app must support OS-level text size scaling (Dynamic Type iOS / Font Size Android) at all system settings without text truncation or layout breakage.

AI Model Quality

- **NFR-M1:** Intent prediction model must achieve ≥ **75% accuracy** on ambiguous-valence messages (message-only, no Soul Document) on the held-out annotated test set before launch. Launch is blocked below this threshold.
- **NFR-M2:** With Soul Document active, intent prediction accuracy must reach ≥ **85%** on the matched message set used for Soul Document ablation testing.
- **NFR-M3:** Model confidence scores must be calibrated — a prediction displayed at 70% confidence must be correct approximately 70% of the time (±5% tolerance). Expected Calibration Error (ECE) ≤ 0.05 before launch.
- **NFR-M4:** The high-ambiguity flag must trigger when max-category confidence is ≤ 55%. Threshold validated on the test set to confirm correct identification of messages human annotators rated as ambiguous (≥ 40% inter-annotator disagreement).
- **NFR-M5:** After 10 user-submitted corrections for a single sender, per-sender calibration must improve accuracy on that sender's messages by ≥ 5 percentage points relative to uncalibrated baseline.
- **NFR-M6:** The I8 pattern detector must achieve ≥ **70% precision** on the I8 test set before the safeguarding signpost is enabled in production. False positive rate ≤ 15% — a false positive signpost is a user harm event.

Scalability

- **NFR-SC1:** The inference API must support **50 concurrent requests** without latency degradation beyond P95 800ms at MVP launch (~500 daily active users, 10% concurrent session assumption).
- **NFR-SC2:** Inference API infrastructure must be horizontally scalable to **500 concurrent requests** through provisioning changes only — no architectural changes required.

- **NFR-SC3:** Local Soul Document storage model must support Soul Document sizes up to **500KB** without performance degradation at NFR-P5 thresholds, accommodating Soul Document v2 (Growth) without a storage migration.
- **NFR-SC4:** PII-redacted correction log system must handle **10,000 correction events per month** without query degradation. Provides 10x headroom at MVP scale.

Integration

- **NFR-I1:** Content script must function correctly across WhatsApp Web DOM versions released within the past **90 days** at any given point. Automated DOM regression tests run on CI against current WhatsApp Web DOM snapshot.
- **NFR-I2:** NC must pin to a specific LLM inference API version. Provider API version upgrades are explicit — no automatic upgrades. Breaking changes to the inference API must not affect production without a validated NC model update.
- **NFR-I3:** QR-code device pairing (FR34) must complete Soul Document transfer within **30 seconds** on a standard home WiFi connection. Transfer is atomic — either complete or rolled back. Partial transfers must not activate.
- **NFR-I4:** Extension must pass Chrome Web Store automated review checks on every release. Mobile app must pass App Store and Play Store automated validation checks on every release. CI pipeline includes store validation checks before release candidate tagging.